BayesINE3. Estimación de medias. Modelo normal-normal

Intro a Estadística Bayesiana

DataLab

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ICMAT-CSIC

Contents

- Chapter objectives
- Basic setup: The normal-normal model
- Basic uses. Inference and prediction
- Bayesian Decision Analysis
- The case with unknown variance: The normal-inverse gamma model
- Final comments
- References
- Lab3

Chapter objectives

Objectives

- Revisit main concepts in Bayesian inference through another simple but very useful model: Estimation of means through the normal-normal model
  - Point estimation
  - Interval estimation
  - Hypothesis testing
  - Prediction
- Consider also the case with unknown variance
- A lab illustrates concepts numerically

Basic setup: The normal-normal model

Basic setup

Problem faced: Estimating a mean (no covariates, this is revised in next session)

A typical example: *Consider detecting fraud in fishing activities for the EnvAg. Only sardines of size bigger than a certain size are allowed for capture by an EU regulation. GC inspectors catch the Valdobeach right at the entrance of Cedeira port. We would like to assess whether the Valdobeach crew has respected the EU regulation.*

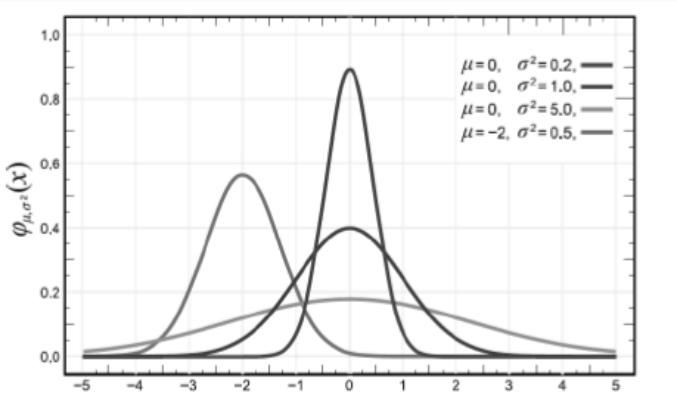
Numerical example in Lab3. Here we go through concepts and methods.

Start with the model. Recall the normal distribution `@https://en.wikipedia.org/wiki/Normal_distribution`

Basic setup

Probabilistic setup

- n observations iid. Continuous measurements compiled as X
- Exploratory analysis suggests normal distribution (possibly after transformation) $X|\theta \sim N(\theta, \sigma^2)$
- Parameter of interest: θ distribution mean; for the moment suppose variance known
- Density: $f(x|\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\theta)^2}{2\sigma^2}\right)$



Basic setup

Classical setup

- Likelihood $l(\theta) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\theta)^2}{2\sigma^2}\right)$
- $\propto \exp\left(-\frac{1}{2\sigma^2}(2\theta \sum_{i=1}^{n} x_i - n\theta^2)\right)$
- log-likelihood $\propto h(\theta) = n\theta^2 - 2\theta \sum_{i=1}^{n} x_i$
- MLE. Maximum likelihood estimator
  - Maximize log-likelihood, $h'(\theta) = 0$
    - $\hat{\theta} = \frac{\sum_{i=1}^{n} x_i}{n} = \bar{x}$

Basic setup

Bayesian setup. θ is random

- Suppose non-informative prior $p(\theta) = 1, \theta \in \mathbf{R}$. Intuition?

Need to aggregate both sources of information: Bayes' formula.
From prior to posterior

- $p(\theta|\mathbf{x}) \propto 1 \times \exp\left(-\frac{1}{2\sigma^2}(2\theta \sum_{i=1}^n x_i - n\theta^2)\right)$

The posterior summarizes all information available about θ.
Corresponds to

$$\theta|\mathbf{x} \sim N(\bar{x}, \sigma^2/n)$$

Sequential update, $\mu_{n+1} = \frac{n\mu_n + x_{n+1}}{n+1} \quad \sigma_{n+1} = \frac{n\sigma_n}{n+1}$

Basic setup

Bayesian setup. Posterior clearly depends on prior....

Suppose a priori

$\theta \sim N(\theta_0, \sigma_0^2\ ) \implies p(\theta) \propto \exp\left(-\frac{1}{2\sigma_0^2}(\theta^2 - 2\theta_0\theta)\right)$

Recall likelihood $\propto \exp\left(-\frac{1}{2\sigma^2}(2\theta\sum_{i=1}^n x_i - n\theta^2)\right)$

Posterior $\propto \exp\left(-\frac{1}{2}(\theta^2(\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}) - 2\theta\left(\frac{\sum x_i}{\sigma^2} + \frac{\theta_0}{\sigma_0^2}\right)\right)$

$\theta|\mathbf{x} \sim N(\mu_n, \sigma_n^2)$ with $\sigma_n^2 = \frac{1}{\left(\frac{n}{\sigma^2}+\frac{1}{\sigma_0^2}\right)}$ and $\mu_n = \frac{\left(\frac{\sum x_i}{\sigma^2}+\frac{\theta_0}{\sigma_0^2}\right)}{\left(\frac{n}{\sigma^2}+\frac{1}{\sigma_0^2}\right)}$

Basic uses. Inference and prediction

Basic uses

Posterior summarises all info available to perform

- Point estimation
- Interval estimation
- Hypothesis testing
- Prediction

'Analytical' results.

But will discuss the use of simulation to provide answers.

Use simulation for all computations also in Lab 3!!!

Basic uses. Point estimation

Summarise the posterior in a single θ value

Posterior mean $\mathbb{E}[\theta \mid \mathbf{x}] = \frac{\frac{n}{\sigma^2}}{\frac{n}{\sigma^2}+\frac{1}{\sigma_0^2}}\,\bar{x} + \frac{\frac{1}{\sigma_0^2}}{\frac{n}{\sigma^2}+\frac{1}{\sigma_0^2}}\,\mu_0$

- As n grows ...

Posterior median. Same

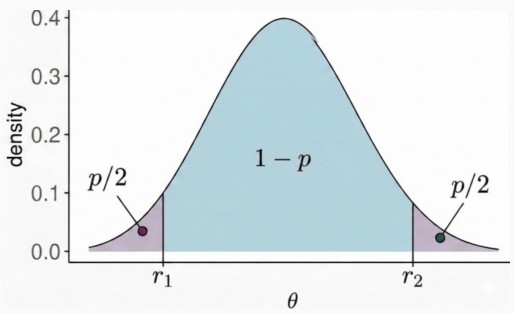
Posterior mode. Same

But choice based on minimising expected loss ...

Basic uses. Interval estimation

Summarise the posterior in an interval with desired probability level $1-p$ (of containing θ). Posterior $N(\mu_n, \sigma_n^2)$

E.g Symmetric interval of probability (1-p)
$[\mu_n - qnorm(1-p/2) \times \sigma_n, \mu_n + qnorm(1-p/2) \times \sigma_n)]$



Other possibilities: HPD Interval (SAME) ,...

Choice based on minimising expected loss...

Basic uses. Hypothesis testing

Which among several hypothesis supported by the evidence?

Testing two bilateral hypothesis $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$
Evidence supports H_0 iff $Pr(\theta \leq \theta_0 | \mathbf{x}) \geq Pr(\theta > \theta_0 | \mathbf{x})$
Evidence supports H_0 iff $Pr(\theta \leq \theta_0 | \mathbf{x}) \geq 1/2$
Evidence supports H_0 iff $pnorm(\theta_0, \mu_n, \sigma_n) \geq 1/2$

Basic uses. Forecasting

Forecasting results in future observations.

Model $X|\theta \sim N(\theta, \sigma^2)$, Posterior $\theta|\mathbf{x} \sim N(\mu_n, \sigma_n^2)$

Predictive distribution

$p(x_{n+1}|\mathbf{x}) = \int p(x_{n+1}|\theta)p(\theta|\mathbf{x})\mathbf{d}\theta$

$X_{n+1}|\mathbf{x} \sim N(\mu_n, 1/(1/\sigma^2 + 1/\sigma_n^2))$

Point forecast μ_n

Interval forecast with probability p

$[\mu_n \pm qnorm(1 - p/2, \mu_n, \sqrt{1/(1/\sigma^2 + 1/\sigma_n^2)})]$

See how uncertainty due to observation σ adds on posterior uncertainty

Bayesian Decision Analysis

Bayesian decision analysis

Decision+State$\rightarrow$ Impact

Assess impact through utility (more, better) or loss (less, better)

Utility (and loss) models also risk attitude

Maximize (Posterior/predictive) expected utility

Testing two bilateral hypothesis $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$

		Is $\theta \leq \theta_0$	. $\theta > \theta_0$
Says	$\theta \leq \theta_0$	1	0
	$\theta > \theta_0$	c	1

Posterior expected utility of saying $\theta \leq \theta_0$ is $Pr(\theta \leq \theta_0|\mathbf{x})$

Posterior expected utility of saying $\theta > \theta_0$,

$c \times Pr(\theta \leq \theta_0|\mathbf{x}) + (1 - Pr(\theta \leq \theta_0|\mathbf{x})$

Say $\theta \leq \theta_0$ if $Pr(\theta \leq \theta_0|\mathbf{x}) \geq \frac{1}{2-c}$

The case with unknown variance: The normal-inverse gamma model

Basic setup

Probabilistic setup

- n observations iid. Continuous measurements
- X compiles the n measurements
- Parameters of interest: θ mean; σ^2 variance
- Exploratory analysis suggests

$$x_1, \ldots, x_n \mid \theta, \sigma^2 \sim \mathcal{N}(\theta, \sigma^2)$$

- The conjugate prior is the Normal–Inverse-Gamma:

$$\theta \mid \sigma^2 \sim \mathcal{N}\left(\mu_0, \frac{\sigma^2}{\kappa_0}\right), \qquad \sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0)$$

Basic setup

The posterior is also Normal–Inverse-Gamma:

$$\theta \mid \sigma^2, x \sim \mathcal{N}\left(\mu_n, \frac{\sigma^2}{\kappa_n}\right), \qquad \sigma^2 \mid x \sim \text{Inv-Gamma}(\alpha_n, \beta_n)$$

with updated parameters:

$$\kappa_n = \kappa_0 + n; \; \mu_n = \frac{\kappa_0 \mu_0 + n\bar{x}}{\kappa_0 + n}; \; \alpha_n = \alpha_0 + \frac{n}{2}$$

$$\beta_n = \beta_0 + \frac{1}{2}\sum_{i=1}^{n}(x_i - \bar{x})^2 + \frac{\kappa_0 n}{2(\kappa_0 + n)}(\bar{x} - \mu_0)^2$$

- μ_n, weighted average of prior mean and sample mean.
- κ_0 acts as a prior sample size.
- α_0 and β_0 encode prior information about the variance.

Basic setup

For a new observation x_{new}:

$$x_{\text{new}} \mid x \sim t_{2\alpha_n}\left(\mu_n,\ \frac{\beta_n}{\alpha_n}\left(1+\frac{1}{\kappa_n}\right)\right)$$

- The predictive distribution is Student-t, reflecting uncertainty in σ^2.

Final comments

Final comments

- Again we have used conjugate priors. The normal-inverse gamma is a conjugate prior of the normal with two parameters (the posterior is also normal-inverse gamma).
  But this is available only in a few models.
- Assessing the prior
  - Noninformative prior. Mean 0, large variance
  - Assess a couple of quantiles and deduce parameters
  - Observables
- Consistency
- Computations (Via simulation)
- Generative models (Machine learning jargon)

References

References

Agresti, Kateri, Grove, Mira (2026) Bayesian Statistics: A primer for data scientists with Python and R.

Hoff (2009). A First Course in Bayesian Statistical Methods.

Lab3

Lab3

The normal-normal model in R